

The Problem

Company

The Houston Parks & Recreation Department's Urban Forestry Division

Context

The Mayor's office is aiming to improve service level agreement (SLA) success rates across all service types for each department.

The Division Manager of Urban Forestry is tasked with reviewing the current SLAs for each public service the division provides and recommending practical key performance indicators (KPIs) to track compliance with current SLAs.

Problem Statement

Perform exploratory data analysis of all Urban Forestry service requests - from 2020 to 2025 - to answer the following business questions:

- At what time of the day are most service requests received?
- What percentage of requests met their SLA each year?
- Are there any patterns or anomalies in request volume each year?
- What are practical key performance indicators (KPIs)?

The Process

Step 1

Create a project design document

- Identify what data sources are available to answer the business questions
- Review the integrity, quality, and limitations of the data

Tools used: Word, Excel

Step 2

Clean raw data from 2020 to 2025

- Clean service request records from the City of Houston's 311's Help Center
- Clean service request records from the Houston Parks & Recreation Department
- Create dimension and fact tables for star schema in PostgreSQL database

Tools used: Python, SQL, DBeaver

Step 3

Create visualizations for insights

- Create visualizations that answer the business questions
- Use visualizations to construct a dashboard

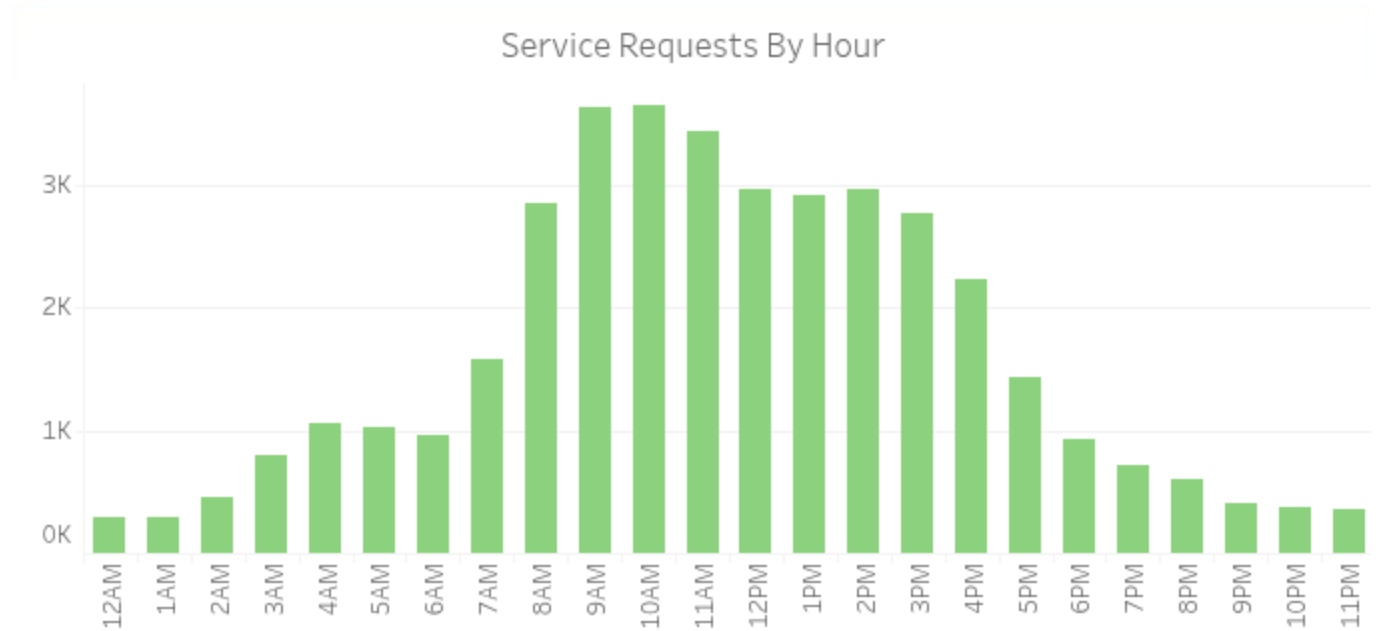
Tools used: Tableau Public

The Results

Highest volume of calls occurs between 9AM and 11AM

Insight:

Service calls are received 24 hours a day. Most calls occur during normal working hours between 6:30AM to 3:30PM, with the highest volume of calls occurring between 9AM and 11AM.



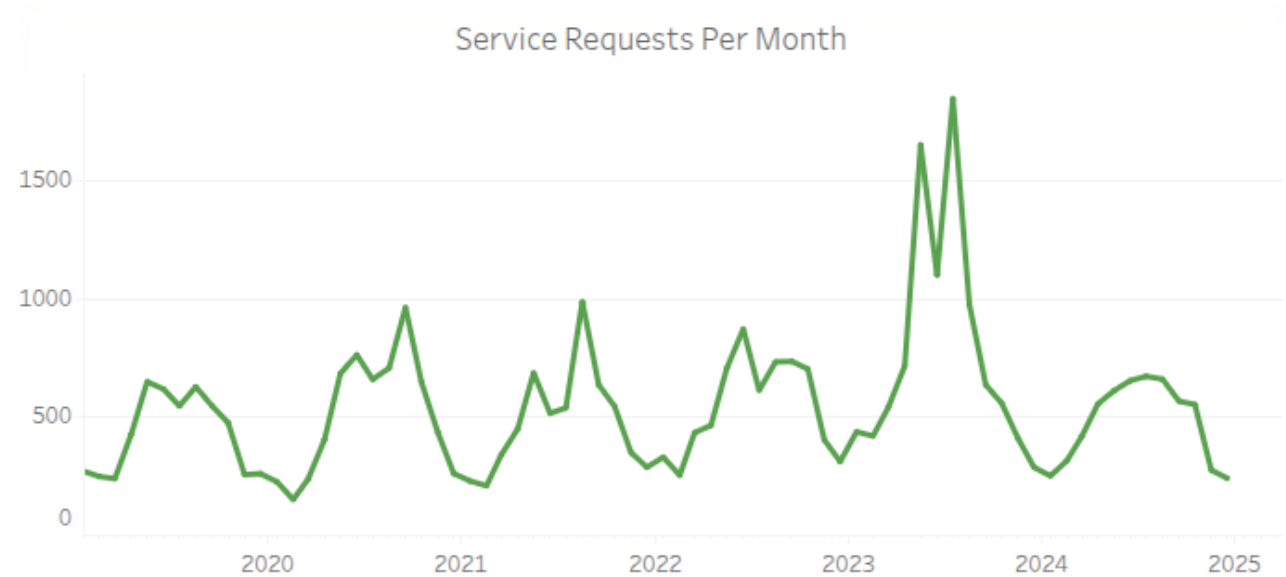
The Results

Request volume is cyclical; peaks March through September each year

Insight:

Annual service request volume is cyclical with a significant increase in requests received mid-March through September. Deciduous trees in Houston typically produce new foliage in mid-March, leading to an increase in service calls for tree maintenance. Volume is lowest during dormant season, October through February.

The unusual spike in service calls observed between 2023 and 2024 is due to two major storm events that were federally declared natural disasters (Derecho and Hurricane Beryl).

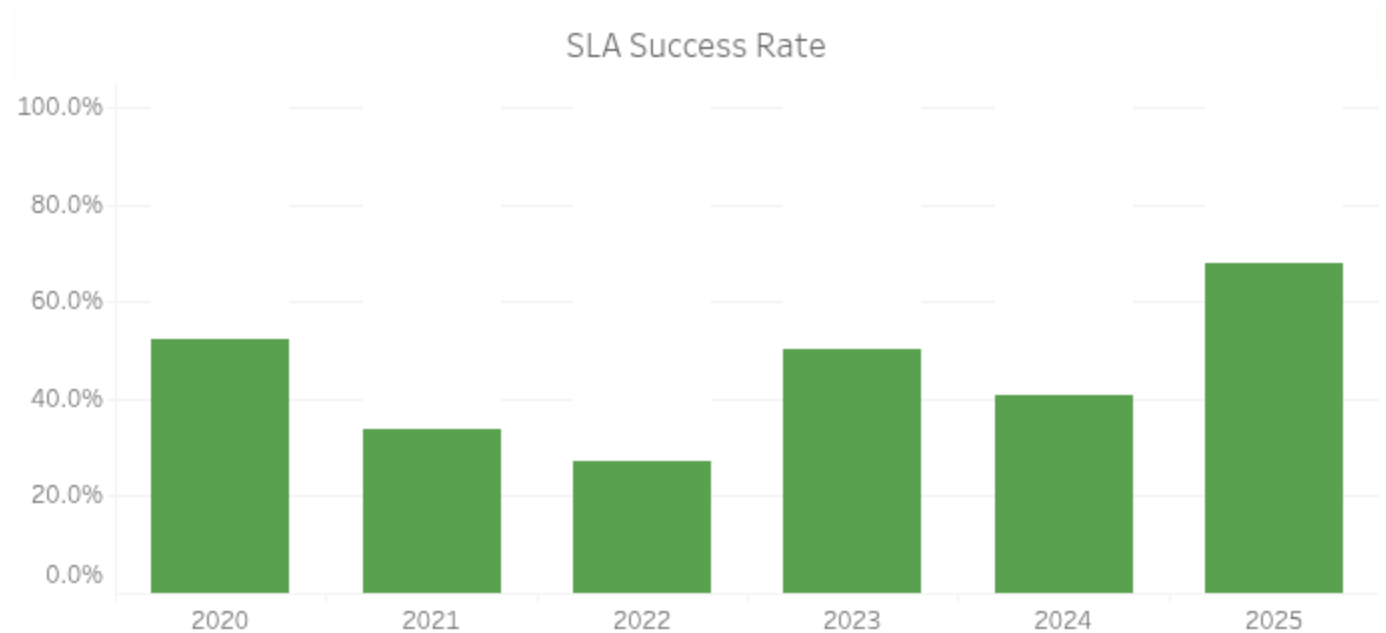


The Results

43% overall SLA success rate over the 5-year period

Insight:

As a whole, SLA success rates have generally been low at 43% over the 5-year period, with the lowest combined success rate of 21% occurring in 2022 and the highest combined success rate of 68% occurring in 2025.



The Results

Higher SLA success rates for critical services; lower rates for non-critical

Insight:

When segmented by the individual service types, SLA success rates have been much higher since 2025 for critical services such as Tree Trimming, and much lower for seasonal and non-critical services such as Tree Planting.

2025 SLA Success Rate	
Tree Code Violation	57%
Tree Planting	33%
Tree Removal	55%
Stump Grinding	62%
Tree Trimming	79%

Recommendation #1

Implement a percentile based SLA and commit to closing 85% of Tier 2

Insight:

Tree Planting consistently had the longest average service time, exceeding 220+ days each year from 2021 to 2024. Requests for Tree Planting are received year-round, but service is only completed from October to February when weather conditions are most optimal for tree survival post-planting. Several extreme outliers also existed for complex tree removals that require utility coordination.

To account for the seasonal nature of tree planting service and outliers from complex tree removals, it is recommended to implement a percentile based SLA and aim to close 85% of Tier 2 each fiscal year.

	Current Median (P50)	Target Tier 1 Good (P75)	Target Tier 2 Standard (P85)	Target Tier 3 Stretch (P90)
Tree Trimming	88	140	175	215
Tree Removal	77	160	220	270
Tree Stump Grinding	119	200	280	360
Tree Planting	119	285	355	415
Tree Code Violation	88	190	235	330

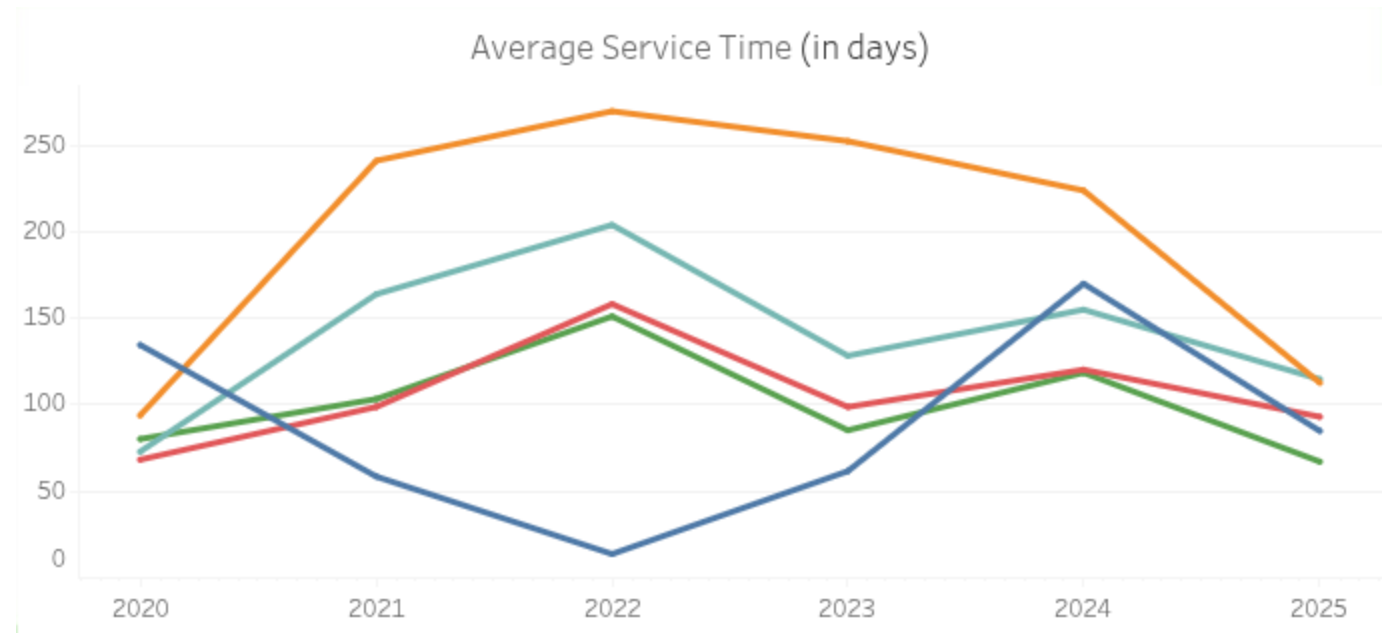
Recommendation #2

Implement KPIs to track average service time by service type

Insight:

Average service times vary widely by service type, having much shorter service times for critical services and much longer service times for seasonal and non-critical services.

Tracking the average number of days it takes to complete each service type, ensuring the average remains below the 85% SLA threshold, will help leadership accurately gauge performance against each SLA.



The Impact

42% higher SLA success rates achieved

Insight:

Only 43% of service requests currently meet the static SLA deadlines. Requests such as Tree Planting drop as low as 10% because the static allotted timeframe of 25 days is unrealistic compared to actual operational timelines (median of 174 days).

Anchoring the SLA success rate to the 85th percentile of historical performance establishes a reasonable and predictable target where only 15% of requests are projected to fail.

This represents a massive 42% increase in overall success rate (increasing from 43% to 85%).

	Current SLA Success Rate	Proposed Target SLA (days)	Expected SLA Success Rate (85%)	Expected SLA Success Rate Increase
Tree Trimming	44.2%	179.5	85.0%	+40.8%
Tree Removal	41.6%	223.5	85.0%	+43.4%
Tree Stump Grinding	47.9%	277.8	84.8%	+36.9%
Tree Planting	10.0%	236.8	83.3%	+73.3%
Tree Code Violation	48.7%	358.2	84.6%	+35.9%